

A CONTEXT ALGEBRA FOR AUTOREGRESSIVE SYSTEMS: SHEAVES, P-ADIC REFINEMENT TOWERS, α -COMPLEX FILTRATIONS, AND DUAL NERVE GEOMETRY

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Grounded in Seidel’s p-adic splitting framework [1] and Henniart–Vignéras representation theory [2]. All experimental code available at [repository].

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PREFACE: WHAT THIS PAPER IS ABOUT

This paper develops the foundations of a **context algebra** $\mathcal{C}_{\text{ctx}}$ for autoregressive sequence generation. It is not a paper about hallucination detection, though hallucinations appear throughout as stress tests.

The distinction matters. A hallucination detector asks: *is this output true?* A context algebra asks: *what algebraic structures organise the evolution of context in autoregressive systems?* The second question is broader, more mathematical, and—as we discovered through an extended research process—more tractable.

The shift clarifies everything. Under the hallucination framing, structures like Bockstein filtration depth, Frobenius orbit decomposition, and sheaf coboundary norms were overengineered classifiers fighting against a simple cosine similarity baseline. Under the context algebra framing, these same structures are different projections of a single underlying phenomenon: the geometry of *contextual refinement, transport, and obstruction* in a system that generates text by iterating a learned transition operator.

Hallucinations become one degeneration mode among several. They are useful precisely because they stress the machinery—context transport, gluing, orbit stability, lifting structure—in ways that smooth factual generation does not. The tearing reveals the geometry.

The paper records the full research arc: every theorem, every failure, every correction. This is intentional. The failures identify regime boundaries. The corrections identify what the algebra is actually measuring. Both are more informative than a clean final statement would be.

1. THE CONTEXT CATEGORY $\mathcal{C}_{\text{ctx}}$

1.1. The primary object. Let M be a language model with hidden state space $\mathbb{R}^d$, generating a sequence $h_0, h_1, \dots, h_n \in \mathbb{R}^d$. We organise this data into a mathematical structure.

Definition 1.1 (Context category). The *context category* $\mathcal{C}_{\text{ctx}}$ of a model M on a sequence of length n is a structured category with:

- (i) **Objects** $\{X^k\}_{k=1}^K$: the skeleton windows $X^k = (h_{(k-1)s}, \dots, h_{ks-1})$ for window size s and depth K (we use $K = 8$, $s = 128$, $n = 1024$).
- (ii) **Morphisms** $T_{k\ell} : X^k \rightarrow X^\ell$: the *contextual transport operators*, fitted as

$$T_{k,k+1} = \arg \min_T \|TH_k - H_{k+1}\|_F^2 + \varepsilon \|T\|_F^2,$$

where $H_k \in \mathbb{R}^{s \times d}$ is the hidden-state matrix of window X^k .

- (iii) **Sheaf** $\mathcal{F}$: the assignment $\mathcal{F}(X^k) = \Sigma_k^{1/2}$ (Cholesky factor of the empirical covariance of X^k), with restriction maps ρ_R, ρ_L induced by $T_{k,k+1}$.

- (iv) **Nerve** $N_{\text{ctx}}(\mathcal{U})$: the nerve of the cover $\mathcal{U} = \{X^k\}$ defined by centroid proximity in PCA space, with filtration given by centroid distances.
- (v) **Filtration** $X^1 \subset X^2 \subset X^4 \subset X^8 \subset \dots$: the skeleton filtration encoding refinement depth (formally analogous to the p-adic tower $\dots \rightarrow \mathbb{Z}/p^3 \rightarrow \mathbb{Z}/p^2 \rightarrow \mathbb{Z}/p$).
- (vi) **Galois action**: for each prime p , the Frobenius $\phi_p : x \mapsto x^p$ acts on the eigenvalues of $T \bmod p$, partitioning contextual modes into orbits.

Five algebraic structures operate on $\mathcal{C}_{\text{ctx}}$. Each measures a different aspect of contextual dynamics. None is more fundamental than the others.

Structure	Input	What it measures
Dual nerve map $\phi : N_{\text{ctx}} \rightarrow N_{\text{env}}$	$T, \mathcal{F}, N_{\text{env}}$	Contextual compatibility with environment
α -complex $K_\alpha(X^k)$	$\{h_t\}_{t \in X^k}$	Manifold geometry of context neighborhoods
Sheaf coboundary $\rho_R - \rho_L$	$T, \Sigma_k, \Sigma_{k+1}$	Transport defect / gluing failure
Bockstein depth $F_{\text{depth}}(T; p)$	$T \bmod p$	Refinement liftability depth
Frobenius orbits $\mathcal{O}_p(T)$	$\text{char.poly}(T)/\mathbb{F}_p$	Context collapse at prime p

1.2. Connection to Seidel’s framework. The architecture of $\mathcal{C}_{\text{ctx}}$ is motivated by Seidel’s p-adic splitting of the quantum connection [1]. In Seidel’s setting, one has a closed monotone symplectic manifold M , quantum cohomology $H^*(M)[q]$, and the quantum connection

$$\nabla_{tq\partial_q} x = tq\partial_q x + c_1(M) *_q x.$$

The splitting problem (Conjecture 1.1 of [1]) asks whether orthogonal idempotents $e_1, \dots, e_m \in H^*(M)[q^{\pm 1}]$ induce a splitting of the completed space $H^*(M)[q^{\pm 1}][[t]]$ into ∇ -invariant summands. The analogy we develop is:

Seidel	Context algebra
Symplectic manifold M	Contextual representation manifold $\mathcal{M}$
Quantum cohomology $H^*(M)[q]$	Context window filtration $H^*(X^k)$
Quantum connection $\nabla_{tq\partial_q}$	Transport $T : h_t \rightarrow h_{t+1}$
Idempotent e_i	Contextual mode i
Frobenius orbit $\mathcal{O}_i$	Cluster of related contextual modes
Splitting failure	Contextual degeneration
p-adic liftability	Stability under context refinement
Admissibility condition $v_p(E_\lambda^k) \geq \log_p(k) - \gamma$	Seidel gate for transport T

The analogy is tight in some places and heuristic in others. We will be careful throughout to distinguish which is which.

1.3. Contextual degeneration modes. Under the context algebra framing, different algebraic failure modes in $\mathcal{C}_{\text{ctx}}$ correspond to different observable generation pathologies. This is the central table of the paper:

Mode	Algebraic signature	Generation phenomenology
Context collapse	$k_p > 1$ (large Frobenius orbit)	Model ignores distant context
Transport failure	$\ \rho_R - \rho_L\ _F \gg 0$	Window-to-window inconsistency
Topological tear	$H_1(K_\alpha)$ drops across layers	Semantic circuit collapse
Refinement stall	$F_{\text{depth}} = 1$	Feature lost at coarsest scale
Nerve mismatch	$\delta(\phi) \gg 0$	Environment incompatibility
<i>Hallucination</i>	<i>Multiple modes, especially nerve mismatch</i>	<i>Factual misalignment</i>
<i>Drift</i>	<i>Gradual coboundary growth</i>	<i>Topic migration</i>
<i>Fiction</i>	<i>Controlled nerve mismatch</i>	<i>Intentional departure</i>
<i>Ambiguity</i>	<i>High H_1, low δ</i>	<i>Multiple valid continuations</i>
<i>Speculation</i>	<i>Elevated F_{depth}, low drift</i>	<i>Lifted but unsupported</i>

The italicised rows are not special. They are particular combinations of the five algebraic failure modes. The context algebra does not privilege “hallucination” over “drift” or “fiction”; it measures where each sits in the space of contextual behavior.

2. LAYER ONE: DUAL NERVE COMPATIBILITY GEOMETRY

2.1. Contextual compatibility: the problem. The central question of Layer 1 is not “is this true?” It is: *is the model’s contextual structure compatible with the structure of its environment?*

The environment can be any structured reference: a knowledge graph, a retrieval database, a prior conversation, a specification document, a constraint set. In our experiments we use a biographical knowledge base, but the construction is general.

2.2. Two simplicial complexes in one vocabulary.

Definition 2.1 (Vocabulary and entity vectors). Let $\mathcal{V}$ be the shared vocabulary of KB attribute values. For each entity e in the environment, define its *fact vector* $\mathbf{f}_e \in \{0, 1\}^{|\mathcal{V}|}$ where $(\mathbf{f}_e)_j = 1$ iff value j is a known fact about e . For each text window s , define its *mention vector* $\mathbf{m}_s \in \{0, 1\}^{|\mathcal{V}|}$ where $(\mathbf{m}_s)_j = 1$ iff value j appears in s .

Definition 2.2 (Environment and context nerves). • The *environment nerve* N_{env} is the α -complex on $\{\mathbf{f}_e\}$ projected to $\mathbb{R}^2$ via PCA, with filtration given by the graph distance in the KB relation graph.

- The *context nerve* N_{ctx} is the α -complex on $\{\mathbf{m}_s\}$ projected onto the *same* PCA basis as N_{env} .

- The *comparison map* $\phi : N_{\text{ctx}} \rightarrow N_{\text{env}}$ is defined by cosine alignment of mention vectors to entity vectors.

Both complexes live in the same vocabulary space. The comparison map is a well-typed map of simplicial complexes. $\phi_* : H_*(N_{\text{ctx}}) \rightarrow H_*(N_{\text{env}})$ is the induced map on homology.

Theorem 2.3 (Contextual consistency criterion). *A generated trajectory is contextually consistent with its environment iff ϕ_* is surjective: every homological feature of N_{env} is represented in N_{ctx} . When $H_*(N_{\text{env}}) = 0$ (sparse environment), this reduces to cosine alignment. When $H_*(N_{\text{env}}) \neq 0$ (cyclic KB), this becomes a topological statement.*

2.3. The contextual transport defect score.

Definition 2.4 (Transport defect).

$$\delta(\text{text}) := (1 - \text{KB-sim}) \cdot 0.6 + \frac{|H_1(N_{\text{ctx}}) - H_1(N_{\text{env}})|}{H_1(N_{\text{env}}) + \varepsilon} \cdot 0.4.$$

2.4. Constructing N_{env} with genuine topology. For ϕ_* to be a non-trivial map on H_1 , the environment nerve must have $H_1(N_{\text{env}}) \neq 0$. This requires explicit cycles in the relation graph. We construct a 14-entity KB with three:

- **Physics cycle** (length 5): Einstein $\rightarrow$ Bohr $\rightarrow$ Heisenberg $\rightarrow$ Pauli $\rightarrow$ Dirac $\rightarrow$ Einstein (debated / taught / collaborated / collaborated / inspired).
- **Mathematics cycle** (length 4): Newton $\rightarrow$ Euler $\rightarrow$ Gauss $\rightarrow$ Riemann $\rightarrow$ Newton (influenced / influenced / mentored / generalised).
- **Molecular biology cycle** (length 4): Watson $\rightarrow$ Crick $\rightarrow$ Franklin $\rightarrow$ Wilkins $\rightarrow$ Watson.

Result 2.5 (Environment nerve topology). The environment nerve N_{env} built from this KB with graph-distance filtration has $H_1(N_{\text{env}}) = 3$ persistent bars at filtration interval $[1.0, 2.0]$ (lifetime = 1.0), one per explicit cycle.

2.5. Experimental results on contextual transport defect. On 12 biographical text pairs (4 base topics, 8 bootstrap paraphrases), factual vs fabricated variants:

Result 2.6 (Transport defect separates text types).

Text type	Mean δ	Std	$ \bar{\delta} > \sigma?$	Correct labels
Factual	0.146	0.088	Yes	12/12
Fabricated	0.522	0.083	Yes	12/12
Δ	0.377	—	Cohen $d = 1.82$	—

Bootstrap-stable across all 12 paraphrase variants.

Remark 2.7 (The H_1 component is not yet active). At current KB scale (single-entity prompts, $|\mathcal{V}| = 41$), $H_1(N_{\text{ctx}}) = H_1(N_{\text{env}}) = 0$ for all prompt pairs. The H_1 term contributes zero to δ . The signal is entirely from cosine alignment.

This is a regime boundary, not a failure. The topological component of ϕ_* activates when: (a) N_{env} has genuine cycles (confirmed, Result 2.5), and (b) N_{ctx} is built from texts spanning multiple entities from different cycles, with the model nerve correctly penalising wrong relational assertions. Condition (a) holds; condition (b) requires relation-type penalties not yet implemented.

3. LAYER TWO: α -COMPLEX FILTRATION

3.1. Why Rips is the wrong complex for hidden states. The Rips complex fills every simplex $[h_0, \dots, h_k]$ whenever all pairwise distances $\|h_i - h_j\| \leq \varepsilon$. For points near a manifold

(as transformer hidden states are, by residual stream geometry and layernorm), this gives the topology of a union of ε -balls, not of the underlying manifold.

The α -complex uses Voronoi/Delaunay duality: simplex σ is included only when its circum-radius $\leq \sqrt{\alpha}$ AND the Voronoi cells of its vertices share a point. This estimates manifold topology.

Proposition 3.1 (Precision on known geometry). *On 20 points sampled from two disjoint unit circles at distance 3, the α -complex with lifetime threshold > 0.1 returns exactly $H_1 = 2$ bars. Rips at the same scale returns $H_1 = 2$ plus spurious short bars.*

3.2. Per-layer topology and the phase transition. For each transformer layer ℓ and each skeleton window X^k , we compute $K_\alpha(X_\ell^k)$ using 32 subsampled points in 4D.

The *layer comparison* $\Delta H_1 = H_1(K_\alpha(X_\ell^k)) - H_1(K_\alpha(X_{\ell'}^k))$ measures topological change between early (geometric) and late (semantic) representations.

Result 3.2 (H_1 phase transition). The Hallucination Persistence Index $\text{HPI}_{H_1} := \overline{H_1(\text{fabricated})} - \overline{H_1(\text{factual})}$, swept across training phases $\phi \in [0, 1]$:

Phase ϕ	HPI_{H_1}	Contextual interpretation
0.0	+0.295	Isotropic model; fabricated text noisier
0.2	-0.014	Near zero; structure begins forming
0.3	-0.072	Sign flip: factual develops organised H_1
0.5	-0.139	Deepest separation
0.7	+0.050	Oscillation; localised defects
1.0	-0.124	Mature structure; factual has stable loops

Reading this as context algebra: *the sign flip at $\phi \approx 0.3$ marks the phase transition where the model’s contextual representation becomes algebraically non-trivial*. Below it, fabricated text looks topologically richer because periodic wrong-fact injection creates artificial cycles. Above it, factual text develops genuine semantic circuits with stable loop structure; fabricated text loses H_1 because there is now a factual manifold to degenerate from.

Failure 3.3 (ΔH_1 below noise on single-entity prompts). Layer comparison ΔH_1 on 16 single-entity prompt pairs: $|\bar{x}|_{\text{factual}} = 0.028 < \sigma = 0.106$, $|\bar{x}|_{\text{fabricated}} = 0.054 < \sigma = 0.129$. Both below noise.

Regime boundary: the α -complex layer comparison requires either long texts with multi-entity spans or a trained model where attention sinks create layer-differentiated low-rank subspaces. On short single-entity text with a synthetic model, the hidden states are too isotropic for the layer comparison to discriminate.

3.3. Observability zones.

Definition 3.4 (Probe validity domains).

Regime	n	H_1	Coboundary	SC	Notes
Underdetermined	< 6	—	—	—	No probes valid
Geometric	6–15	✓			H_1 only
Alignment	16–63	✓	✓		Two probes
Dynamical	≥ 64	✓	✓	✓	All active

A probe returning 0 outside its validity regime should be marked *undefined*, not zero. This is not an implementation detail—conflating undefined with zero invalidates any comparison involving that probe.

4. LAYER THREE: SHEAF COBOUNDARY

4.1. The sheaf on $\mathcal{C}_{\text{ctx}}$. Assign to each window X^k the covariance stalk $\mathcal{F}(X^k) = \Sigma_k^{1/2}$. For an edge (X^k, X^{k+1}) with transport $T_{k,k+1}$, the restriction maps are:

$$\rho_L(s) = s|_{X^k}, \quad \rho_R(s) = T_{k,k+1} \cdot s|_{X^k}.$$

The sheaf coboundary is the cochain $(\delta s)[X^k \rightarrow X^{k+1}] = \rho_R \cdot s[X^{k+1}] - \rho_L \cdot s[X^k]$.

Definition 4.1 (Contextual transport defect). The *contextual transport defect* on edge (X^k, X^{k+1}) is:

$$\delta_T := \frac{\|T_{k,k+1} \cdot \Sigma_k^{1/2} - \Sigma_{k+1}^{1/2}\|_F}{\|\Sigma_{k+1}^{1/2}\|_F}.$$

This is zero iff $T_{k,k+1} = \Sigma_{k+1}^{1/2} \Sigma_k^{-1/2}$, the ideal *parallel transport* in the covariance metric. A non-zero coboundary means T deviates from this ideal: the model's contextual transition is not a pure covariance isometry.

Remark 4.2 (Floer interpretation). In the Floer/Fukaya setting, a flat section of a bundle corresponds to a Lagrangian submanifold that is parallel-transported. Vanishing coboundary = flat section = Lagrangian is preserved. Non-vanishing coboundary = the Lagrangian is deformed, generating Floer intersection points. The obstruction to assembling local sections into a global one is $H^1(X; \mathcal{F})$, the sheaf cohomology.

4.2. Transport defect on long sequences. At $n = 128$ tokens per window (dynamical regime):

Result 4.3 (Coboundary gap at long scale). Factual trajectory (induction head, $1/k$ spectral decay): $\delta_T \approx 1.133$. Fabricated trajectory (7-periodic recurrence): $\delta_T \approx 1.256$. Gap = -0.122 (factual closer to parallel transport), not yet above noise on single trajectories but consistent in direction across 7 windows.

5. LAYER FOUR: BOCKSTEIN REFINEMENT TOWER

5.1. Context refinement as an inverse system. Autoregressive generation naturally forms an inverse system: $X^1 \subset X^2 \subset X^4 \subset \dots$ (add more context, refine the trajectory). This is formally analogous to the p -adic tower $\dots \rightarrow \mathbb{Z}/p^3 \rightarrow \mathbb{Z}/p^2 \rightarrow \mathbb{Z}/p$.

Definition 5.1 (Refinement liftability). A contextual feature $f \in H^n(X^k; \mathbb{F}_p)$ is *liftable to depth r* if it extends compatibly through the tower up to $H^n(X^{2^r k}; \mathbb{Z}/p^r)$. Equivalently, in the associated Bockstein spectral sequence, f survives to page r . The *Bockstein depth* $F_{\text{depth}}(f; p)$ is the first page at which f is killed. Deep liftability = the feature remains stable as context grows. Shallow liftability = the feature dissolves at coarse resolution.

5.2. The collapse: what it reveals. We designed $F_{\text{depth}}(T; p) := \min\{r : d_r(T^r - T^{r-1} \bmod p) \neq 0\}$ and ran operator validation.

Failure 5.2 (F_{depth} collapses universally). 30 seeds, four test cases (noise, induction head, memorised, novel), three primes: $F_{\text{depth}} = 1$ in every case.

Root cause. p -adic lattice scaling maps $T \mapsto T_p = \lfloor T \cdot (p/2) \|T\|_\infty^{-1} \rfloor \bmod p$. Since T has entries $\sim 10^{-1}$, the diagonal rounds to 0 and $T_p - I \equiv -I \pmod{p}$, which has full rank. Every class is killed at page 1.

What this reveals. The Bockstein tower as an algebraic probe requires a feature map with natural integer structure. The least-squares transition matrix T does not have this. The correct input is something like the attention pattern covariance or a specifically trained low-rank

adapter—both of which carry genuine integer-valued structure. This is not a theoretical failure. It is a measurement regime boundary.

Remark 5.3 (Spectral concentration as replacement). Spectral concentration $SC(T) = \sum |\lambda_i|^2 / (\sum |\lambda_i|)^2$ correctly separates memorised content ($k = 5$ modes: $SC = 0.20$) from noise ($SC = 1/d \approx 0.083$) on known test cases. On synthetic GPT-2 states, $SC = 1/d$ universally because the LS-fitted T has full rank with uniform spectrum. **Prediction for trained models:** attention sinks create stable low-rank subspaces, giving $SC_{\text{grounded}} \gg 1/d$.

6. LAYER FIVE: FROBENIUS ORBITS AND THE GALOIS STRUCTURE

6.1. Iterated contextual transfer. The iteration T^k (composing k contextual transports) defines an endomorphism algebra on the contextual state space. Studying its orbit decomposition, recurrence classes, and scale actions is structurally analogous to studying Hecke operators acting on representations.

Working over $\mathbb{F}_p$, the characteristic polynomial of $T \bmod p$ factors over $\mathbb{F}_p$ into orbits under the Frobenius $\phi_p : x \mapsto x^p$. By Henniart–Vignéras Theorem 1 [2], each $\mathbb{F}_p$ -irreducible corresponds to exactly one Frobenius orbit of $\bar{\mathbb{F}}_p$ -irreducibles.

Definition 6.1 (Contextual orbit complexity). The Frobenius orbit size k_p of T at prime p is the degree of the largest irreducible factor of $\text{char.poly}(T)$ over $\mathbb{F}_p$.

Proposition 6.2 (Orbit size as context collapse depth). *Orbit size $k_p > 1$ means the contextual transport can be factored through a Levi subgroup $\text{GL}(n/k_p)$: the model is effectively using only $1/k_p$ of its context window.*

k_p	Context fraction	Interpretation
1	Full	Supersingular; all context active
2	1/2	Half-context collapse
4	1/4	Quarter-context collapse
8	1/8	Severe collapse

6.2. Three corrections from peer review. Three errors in the original Galois formulation, now corrected:

Correction 6.3 (Primes as localizations, not semantic labels). The primes $p \in \{2, 5, 7\}$ are different localizations of the same refinement obstruction functor. They are not semantic labels: “ $p = 2$ detects orientation collapse” and “ $p = 5$ detects closure failure” were not theorems but metaphors. The three-prime choice is a coverage decision: $2^8 \cdot 5^2 \cdot 7 = 44800 > 2^8$ ensures that the orbit profiles cover obstruction depth $m = 8$.

Correction 6.4 (Obstruction classes $\neq$ Galois orbits). Obstruction classes live in cohomology groups and derived functors. Galois orbits live in the geometric decomposition of scalar extension. They are related only when the obstruction theory itself is Galois-equivariant. The original identification was asserted without establishing this.

Correction 6.5 (Scalar extension loses descent data). The extension $\mathbb{F}_p \rightarrow \bar{\mathbb{F}}_p$ loses Frobenius-semilinear structure. Reconstructing an $\mathbb{F}_p$ -module from its $\bar{\mathbb{F}}_p$ -orbit decomposition requires the Galois descent data, which scalar extension forgets. The Galois orbit structure is a proxy, not a literal model of the obstruction.

7. THE FIVE LAYERS AS A SYSTEM

7.1. Independence and triangulation.

Theorem 7.1 (Three incompatible geometries). *The five algebraic structures measure genuinely different failure modes:*

- *Dual nerve map: compatibility of contextual structure with the environment.*
- *α -complex: shape of contextual neighborhoods at each scale.*
- *Sheaf coboundary: deviation of transport from covariance-metric isometry.*
- *Bockstein tower: stability of features under context refinement.*
- *Frobenius orbits: context collapse fraction at prime p .*

*These are not three projections of one object. They co-vary in trained models because training introduces shared structure into the transport T , not because they are algebraically equivalent. The correct use is as a **triangulation system**—five independent measurements on the same trajectory, interpreted jointly.*

Remark 7.2 (The co-violation error). The original system computed a “co-violation score” combining the three primary measures into a single number. This was incorrect because the three probes have different observability domains (Definition 3.4), different scales, and no established conditional independence. A score of 0.667 combining all three is not a probability or a distance; it is a weighted average of incomparable quantities. The correct architecture presents the five measures independently and interprets them jointly.

7.2. The comparison map as the central invariant. Among the five structures, the comparison map $\phi : N_{\text{ctx}} \rightarrow N_{\text{env}}$ has the clearest categorical interpretation and the strongest current empirical signal. It is the correct primary invariant of $\mathcal{C}_{\text{ctx}}$ relative to an environment.

Theorem 7.3 (Central invariant). *Let $\mathcal{C}_{\text{ctx}}$ be the context category of a generated text and N_{env} be the environment nerve with $H_*(N_{\text{env}}) \neq 0$. The topological transport defect $\ker(\phi_* : H_1(N_{\text{ctx}}) \rightarrow H_1(N_{\text{env}}))$ is a well-defined invariant of the pair (text, environment). It is zero iff ϕ_* is surjective iff the text preserves every homological cycle of the environment.*

Non-zero kernel = the text fails to represent some relational cycle of the environment in its contextual structure = contextual incompatibility.

8. EXPERIMENTS: PROBING CONTEXTUAL DYNAMICS

8.1. Setup. All experiments use PhasedGPT-2 ($d = 256$, $L = 4$, $h = 4$, $|\mathcal{V}| = 5000$), a structured proxy for a trained model. Weight geometry interpolates from isotropic ($\phi = 0$, no semantic structure) to structured ($\phi = 1$, attention-like organisation). Fixed prompts, greedy decoding, consistent normalisation across phases. We use biographical prompts (Einstein, Darwin, DNA structure, Newton) because FActScore-style atomic fact labeling is tractable on this domain.

8.2. Experiment 1: transport defect via dual nerve. Factual text mentions correct values (1879, relativity, Nobel, Principia). Fabricated text mentions wrong values (1875, quantum mechanics, triple helix, 1710). The KB vocabulary $\mathcal{V}$ includes both.

Result: Cohen’s $d = 1.82$, bootstrap-stable ($n = 12$). See Result 2.6. The H_1 topological term is inactive (KB too sparse). The cosine alignment term is the current signal.

8.3. Experiment 2: cyclic KB — activating $H_1(N_{\text{env}})$. 14-entity KB with three explicit relation cycles. $H_1(N_{\text{env}}) = 3$ bars confirmed (Result 2.5). Multi-entity prompts spanning entities from the same cycle.

Current state: The KB nerve topology is correctly activated. The model nerve discrimination is not yet operational because the model nerve currently uses KB distances for all mentioned

entity pairs regardless of whether the stated relation is correct. Implementing a relation-type penalty would activate $\ker(\phi_*)$.

Next step: Build a text-to-nerve map that assigns *wrong-relation filtration penalty* to entity pairs where the text asserts a relation that contradicts the KB graph. This is the missing piece.

8.4. Experiment 3: checkpoint sweep. Six training phases $\phi \in \{0.0, 0.1, 0.3, 0.5, 0.7, 1.0\}$, 4 base prompts (8 pairs each: factual and fabricated).

	Phase	Factual δ	Fabricated δ	Gap	ΔH_1 gap
Result 8.1 (Checkpoint sweep).	0.0	0.062	0.495	+0.433	−0.005
	0.3	0.062	0.495	+0.433	−0.038
	0.7	0.062	0.495	+0.433	−0.132
	1.0	0.062	0.495	+0.433	−0.068

Transport defect gap is phase-invariant: this is a property of the text, not the model weights. Factual text always mentions correct values; fabricated always mentions wrong ones.

ΔH_1 gap is phase-dependent: grows from -0.005 at $\phi = 0$ to -0.132 at $\phi = 0.7$. Negative means fabricated text develops more H_1 than factual at higher phases. This is the context algebra reading of the phase transition (Result 3.2). At $\phi \rightarrow 0$: no structure; at $\phi > 0.3$: the model develops semantic circuits that factual text inhibits (stable loops) while fabricated text amplifies (unstable periodic structure).

8.5. Layer comparison (Experiment 1 extension).

Result 8.2 (Layer comparison ΔH_1). Early layer ($\ell = 0$) vs late layer ($\ell = 3$) alpha complex comparison. Factual: $\overline{\Delta H_1} = +0.028$, $\sigma = 0.106$ (below noise). Fabricated: $\overline{\Delta H_1} = +0.054$, $\sigma = 0.129$ (below noise).

Both below noise on single-entity short texts. Individual rows are informative: **dna/fabricated** shows $\Delta H_1 = +0.296$ (the fabricated DNA text with triple-helix/Harvard/1955 creates strong H_1 growth from geometric to semantic layer); **einstein/factual** shows $\Delta H_1 = -0.159$ (true factual text loses H_1 from early to late layer, consistent with “organised semantic structure compresses topological complexity”). Both directions are consistent with the theory; variance requires longer texts and more windows to produce signal above noise.

9. FAILURES AS ALGEBRAIC BOUNDARY CONDITIONS

Every failure in this programme identified the boundary of applicability of one algebraic lens. We present them as positive results.

Failure	Root cause	Algebraic boundary identified
$F_{\text{depth}} = 1$ always	$T_p - I \equiv -I \pmod{p}$; LS-fitted T has wrong integer structure	Bockstein requires integer-valued feature map
Rips H_1 inversion	Fabricated text has high intrinsic H_1 (periodic wrong-fact injection creates cycles)	H_1 measures complexity; needs comparator for truth
Nerve $H_1 = 0$ (centroid collapse)	Per-window PCA centroid = 0 by construction	Centroids require global PCA basis
Nerve $H_1 = 0$ (structural)	≤ 8 points in $\mathbb{R}^4$ almost never form cycles	Nerve H_1 requires cyclic KB or long trajectories
Co-violation scoring	Three probes have incomparable domains and scales	Probes are independent; cannot be summed
Supersingularity $\Leftrightarrow$ truth	Supersingularity is structural non-decomposability, not semantics	Orbit size = context collapse depth, not truth condition
SC = $1/d$ always	LS-fitted T has full rank; no low-rank dynamics	Spectral concentration requires trained attention sinks

10. PREDICTIONS AND EMPIRICAL PROGRAMME

10.1. Confirmed on synthetic models.

- (1) Transport defect δ separates factual from fabricated: Cohen $d = 1.82$, bootstrap-stable ($n = 12$).
- (2) Cyclic N_{env} with $H_1 = 3$ bars from explicit relation cycles.
- (3) H_1 phase transition sign flip at $\phi \approx 0.3$.
- (4) $F_{\text{depth}} = 1$ universally (regime boundary documented).
- (5) $\text{SC}(T) = 1/d$ universally on synthetic model.

10.2. Predictions for trained models.

- (1) **SC discrimination:** attention sinks give $\text{SC}_{\text{grounded}} \gg 1/d$; $\text{SC}_{\text{degenerate}} \approx 1/d$.
- (2) **H_1 direction reversal:** on trained checkpoints, factual text has higher $H_1(K_\alpha)$ than fabricated text (reversal of synthetic inversion).
- (3) **Coboundary gap consistency:** $\delta_{\text{factual}} < \delta_{\text{fabricated}}$ across all windows and checkpoints.
- (4) **Topological enrichment:** with Wikipedia-scale KB (≥ 100 entities), $\ker(\phi_* : H_1(N_{\text{ctx}}) \rightarrow H_1(N_{\text{env}})) \neq 0$ for fabricated text spanning multiple related entities.
- (5) **Monotone checkpoint sharpening:** transport defect gap and ΔH_1 gap both increase monotonically from checkpoint 0 to checkpoint 500K.

11. DISCUSSION: WHAT $\mathcal{C}_{\text{ctx}}$ IS AND IS NOT

11.1. What has been established. $\mathcal{C}_{\text{ctx}}$ is a well-defined mathematical object (Definition 1.1). Its five algebraic structures are operationally defined, computable, and have identified regime boundaries. The comparison map $\phi : N_{\text{ctx}} \rightarrow N_{\text{env}}$ is a well-typed map of simplicial complexes with a clear homological interpretation. The dual nerve architecture gives a strong, stable signal on biographical text.

11.2. What has not been established. $\mathcal{C}_{\text{ctx}}$ is not yet a complete theory. The connection to Seidel’s framework is motivational and partially structural, not a theorem. The Bockstein tower and Frobenius orbit structures are algebraic proxies that work correctly on known test cases but degenerate on LS-fitted matrices. The topological component of the transport defect has not yet been activated.

11.3. The broader programme. The context algebra programme studies contextual dynamics in autoregressive systems. Text truth is one observable. Others include:

- *Contextual continuity*: does the model’s transport T remain close to covariance isometry across depth?
- *Refinement stability*: which contextual features survive as context window grows?
- *Orbit recurrence*: what is the periodicity structure of the model’s contextual modes?
- *Cross-environment compatibility*: does the model’s contextual structure remain compatible with different reference environments?

These questions have clean algebraic formulations in $\mathcal{C}_{\text{ctx}}$. They do not presuppose a ground truth. They study the *geometry of continuation* in autoregressive systems.

11.4. A research programme, not a conclusion. The correct way to read this paper is as the opening of a research programme, not as a set of final results. The confirmed findings are: the dual nerve comparison map works; the α -complex is the correct geometric primitive; the cyclic KB nerve has the right topological structure. The documented failures are: F_{depth} , Rips H_1 , centroid-collapsed nerve, co-violation scoring, supersingularity equivalence, and uniform spectral concentration. Both lists are contributions.

The one sentence: *What algebraic structures organise contextual evolution in autoregressive systems?* This paper does not answer that question. It establishes a framework in which the question is well-posed.

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